Yeming Dai

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RESEARCH INTERESTS

Reliable LLM agents; verification and evaluation; human oversight and state tracking in long-running agent workflows; robust machine learning.

EDUCATION

Arizona State University | Jan 2024-Dec 2026 (expected)

B.S. in Computer Science (Software Engineering) | GPA: 4.0/4.0

Northern Arizona University | Non-degree physics coursework | Dec 2025-Jan 2026

Anhui University of Science and Technology | Sep 2021-Jul 2023

Software Engineering undergraduate studies; no degree awarded

PUBLICATIONS AND MANUSCRIPTS

Adrian Ciotinga, **Yeming Dai**, and YooJung Choi. *PeTeR: Post-Training Robustification of Probabilistic Circuits.*

Workshop paper: UAI 2026 Workshop on Tractable Probabilistic Modeling (published). **Extended manuscript:** AISTATS 2027 (under review).

RESEARCH EXPERIENCE

Undergraduate Researcher in Probabilistic Models

Arizona State University | *Mar 2026-Present*

Research on post-training robustness of probabilistic circuits under distribution shift.

- Diagnosed probability-normalization errors in research code and validated corrected updates while retaining gradient propagation.
- Investigated evaluation confounds through controlled reruns with fixed labels, ruling out label mismatch as the cause of poor clustering.
- Executed and audited 3,360 hyperparameter configurations and a separate GPU study across 28 datasets; reconstructed experiment environments and selected concurrency from measured GPU use.
- Implemented probabilistic-model training for mixed-feature data with missing-value handling; prepared reproducible datasets with fixed splits and cross-validation indices.
- Cross-checked manuscript descriptions, result tables, and cited sources; prepared suggested revisions for clarity and consistency.

Further details on ongoing work will be shared when publication and lab confidentiality requirements permit.

INDEPENDENT SYSTEMS WORK

LLM Agent Workflows and Independent Review

Independent project | *2026* | Method | Prototype | Technical note (non-peer-reviewed)

- Separated stable requirements from changing task state, then distinguished accepted results, current tasks, and conditional next steps when future plans proved dependent on execution outcomes.
- Implemented automatic execution-review iterations with separate agent contexts, evidence-linked decisions, checkpoints, and recovery after interruptions.

LLM Agent Workflows and Independent Review continued

- Added operator tools for state inspection and human approval; applied the workflow to reviewed software changes and multilingual transcription checks.
- In a packaging task, independent agent review caught a symbolic-link loophole despite passing tests; subsequent iterations repaired the directory checks.
- Compared candidate claims with prior work and narrowed subsequent questions to verification behavior and human decision boundaries.

SELECTED TECHNICAL PROJECTS

Classroom Live Transcriber

Independent developer | *May-Oct 2026* | Repository

- Built an on-device Apple Silicon lecture-transcription application using Metal-accelerated whisper.cpp, overlapping audio chunks, and asynchronous workers.
- Preserved raw and cleaned transcripts with session configuration; implemented verified model downloads, atomic file publication, and failure-safe retries.
- Added independent audio-language selection, session reset and incremental copying, plus no-overwrite transcript renaming and full-text recovery after file loss.
- Released version 1.1.0 with 297 regression tests; verified build-source identity, the packaged application, and the published download artifact.

AudioShifter

Independent developer | *Mar 2025-Aug 2026* | Repository

- Independently developed the original Windows application and an Apple Silicon macOS version; delivered the tested, packaged macOS pre-release within 3 days in August 2026.
- Implemented a modular architecture with background processing, cancellation, child-process cleanup, and source-file protection.
- Verified the pre-release with 165 automated tests, including fault injection, real audio pipelines, packaged execution, and GUI behavior; recorded no failures or skips.
- Audited 75 native binaries and 324 dynamic-library references, then validated the packaged application on a separate non-development Mac.

HONORS AND SCHOLARSHIPS

New American University Transfer Scholarship, ASU (four semesters).

Dean's List, ASU: Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026.

TECHNICAL SKILLS

Programming: Python, Java, JavaScript, C/C++, Bash/Shell

ML and research computing: PyTorch, CUDA, Triton, PyJuice, NumPy, Slurm

Systems and tooling: Linux, macOS, Git, pytest, Metal, whisper.cpp, PySide6, FFmpeg, PyInstaller